COMPUTER SCIENCE DISSERTATION

VIDEO PRESENTATION QUESTION AND ANSWER

Assessment Type	Weight	Requirements
Dissertation	75.00	15,000 word dissertation (excluding appendices) in printed in electronic form (PDF). Any source code must also be included with the electronic submission.
Practical	15.00	Demonstration of project results.
Report	10.00	Project plan including ethics clearance.

Practical

Recorded video presentation (10%) covering the **background, motivation, aims, objectives, workplan, software, results, and impact** of the final year project.

Q and A session (5%) with a panel of 3 where students respond to general questions about their project.

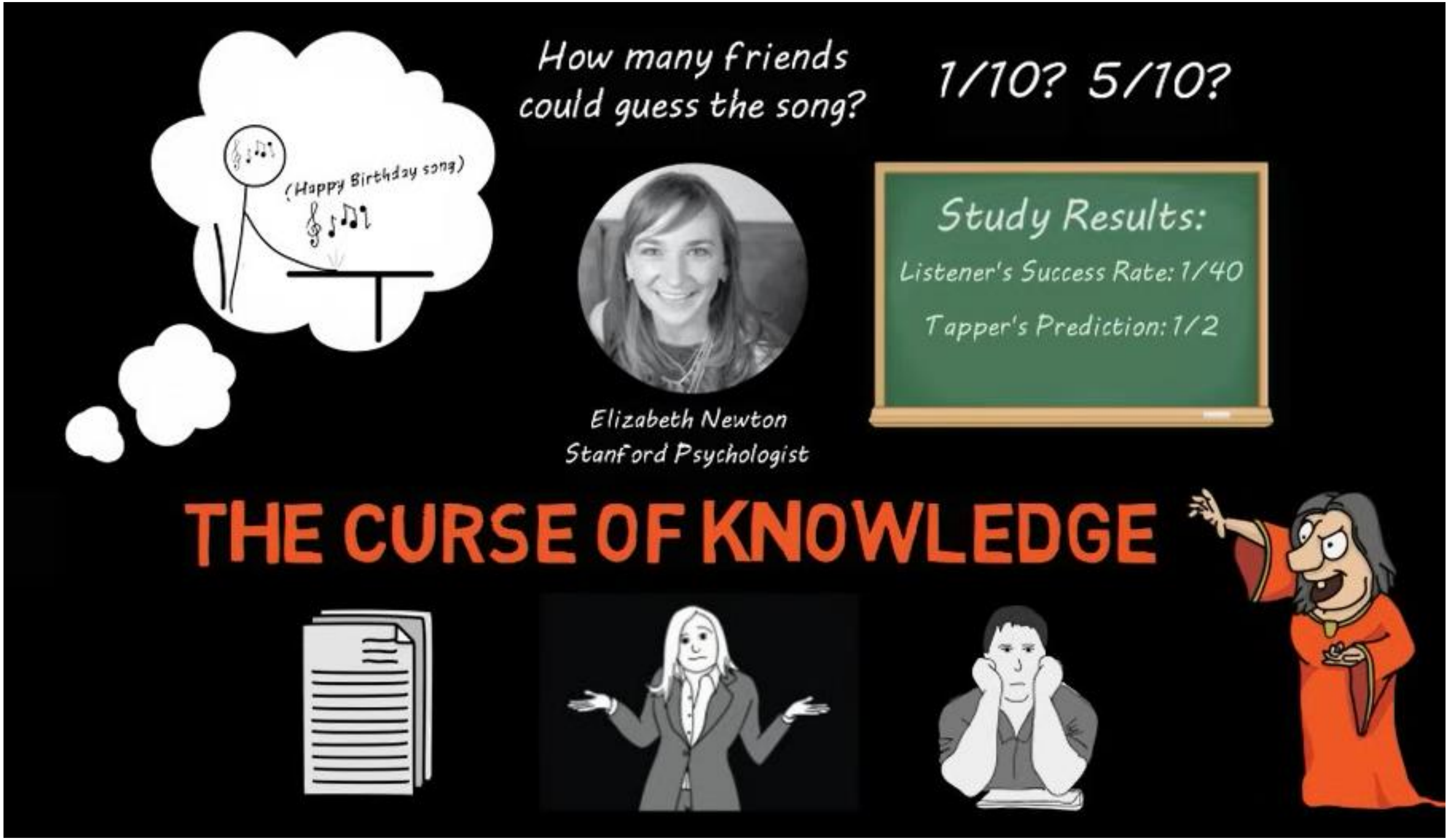
Assessment criteria

Knowledge and Understanding: Individual projects may arise from any area of the curriculum and so may address any of the knowledge and understanding outcomes.

Intellectual Skills: The ability to think independently while giving due weight to the arguments of others. The ability to understand complex ideas and relate them to specific problems or questions. The ability to analyse systematically and effectively, substantial quantities of information.

Professional Skills: Enhanced programming abilities. The ability to comprehend and apply software engineering methodologies.

Transferable Skills: The ability to plan, organisation and execute project work. The ability to communicate their results to others.

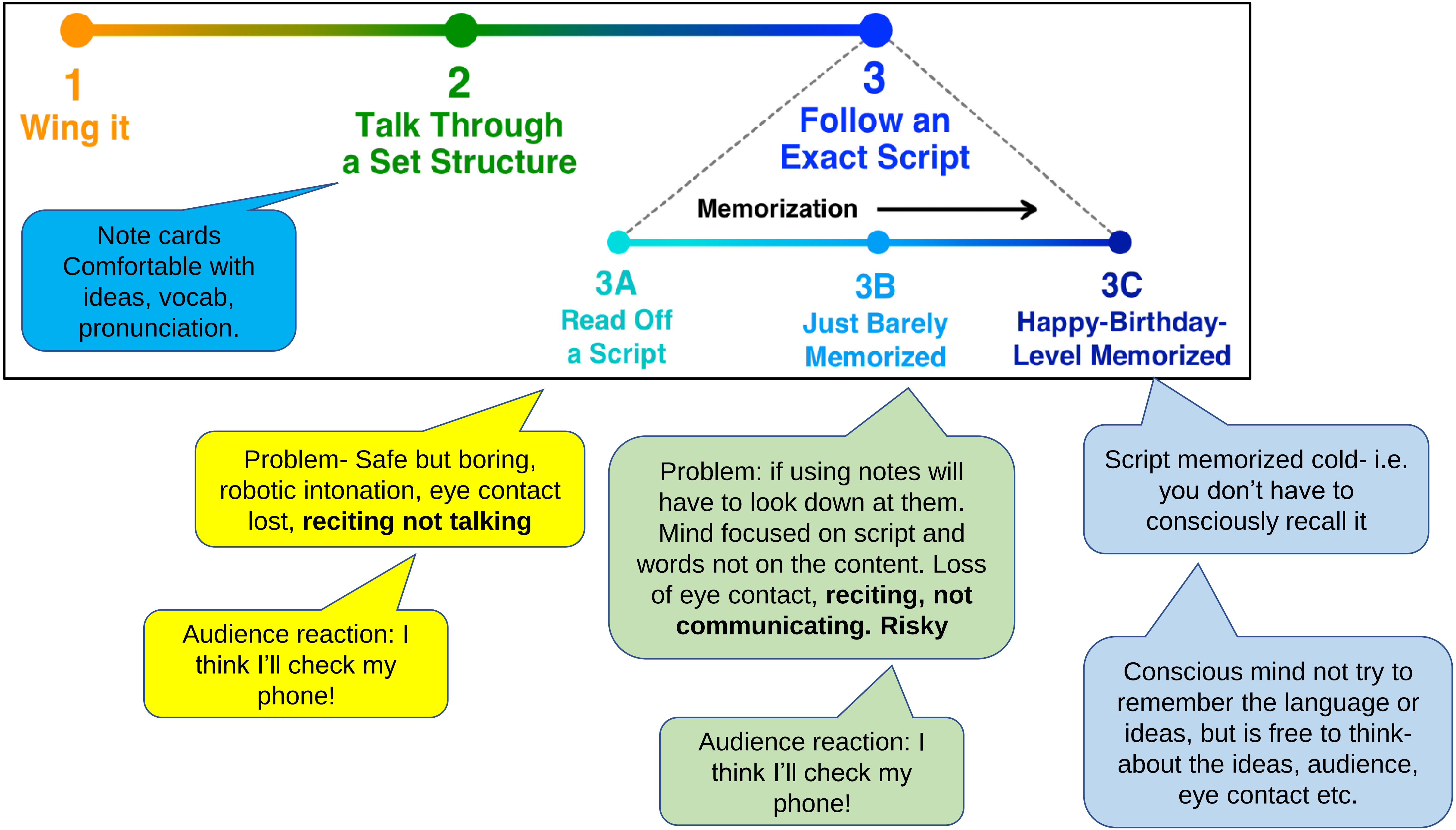

How many friends could guess the song?
1/10? 5/10?
(Happy Birthday song)
Elizabeth Newton
Stanford Psychologist
Study Results:
Listener's Success Rate: 1/40
Tapper's Prediction: 1/2
THE CURSE OF KNOWLEDGE

The **curse of knowledge** is a cognitive bias that occurs when an individual, who is communicating with others, assumes that others have information that is only available to themselves, assuming they all share a background and understanding. This bias is also called by some authors the **curse of expertise**

Software engineering and user interfaces

Software engineers create user interfaces that they themselves can understand and use, but end users - who do not possess the same level of knowledge - find the user interfaces difficult to use and navigate.

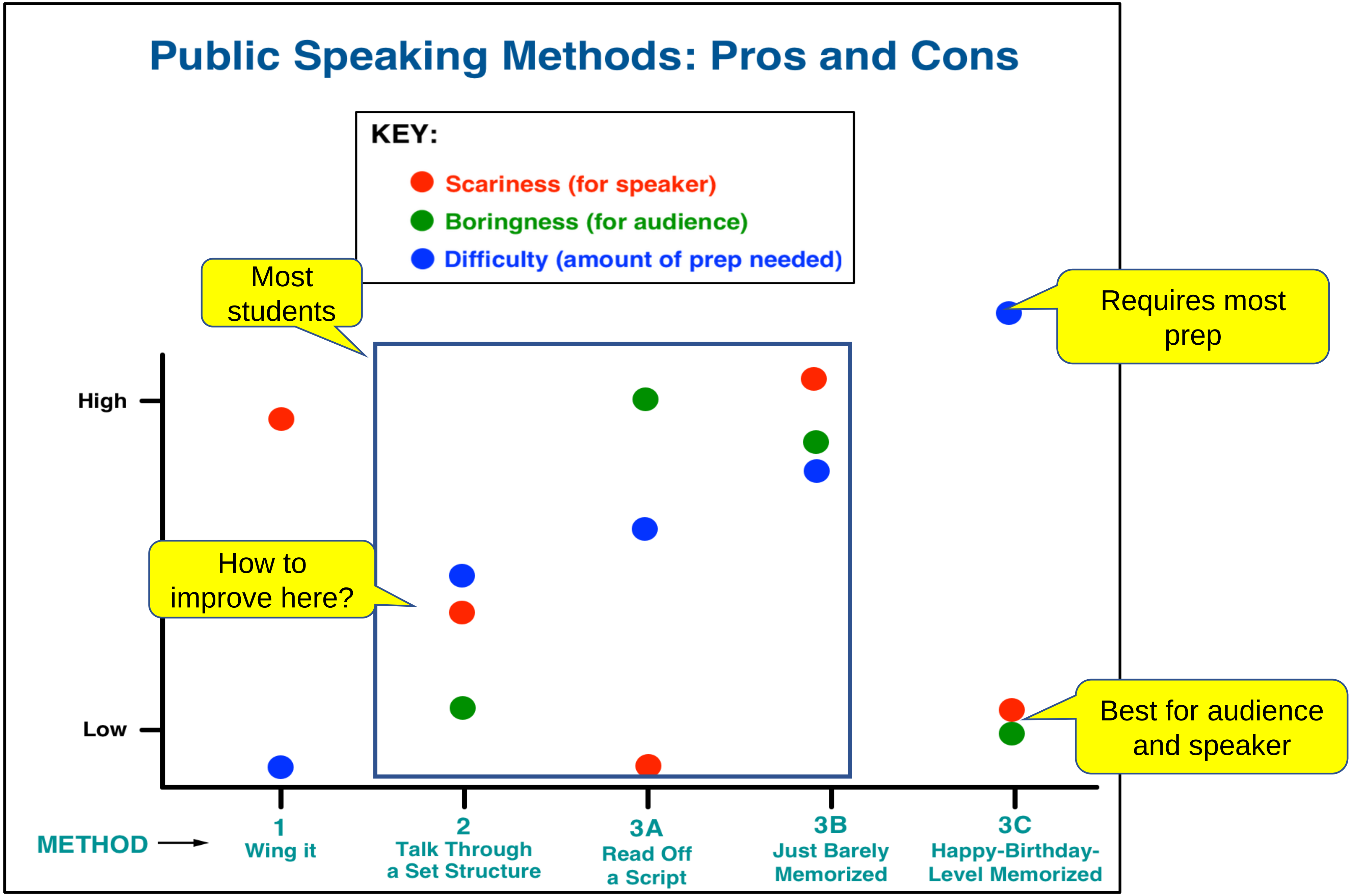
The phrase **"You are not the user"** has become ubiquitous in the user experience industry to remind practitioners that their knowledge and intuitions do not always match those of the end users they are designing for.


1
Wing it
2
Talk Through a Set Structure
3
Follow an Exact Script
Memorization
3A
Read Off a Script
3B
Just Barely Memorized
3C
Happy-Birthday-Level Memorized
Note cards Comfortable with ideas, vocab, pronunciation.
Problem- Safe but boring, robotic intonation, eye contact lost, **reciting not talking**
Audience reaction: I think I'll check my phone!
Problem: if using notes will have to look down at them. Mind focused on script and words not on the content. Loss of eye contact, **reciting, not communicating. Risky**
Audience reaction: I think I'll check my phone!
Script memorized cold- i.e. you don't have to consciously recall it
Conscious mind not try to remember the language or ideas, but is free to think- about the ideas, audience, eye contact etc.

The people at TED refer to two tests you need to pass to qualify as Happy-Birthday-level memorized:

1. If you record yourself saying the talk and play it back at 2x speed, can you say it out loud while it's playing and stay *ahead* of the recording?

2. Can you recite the talk with no problem while simultaneously doing an unrelated task that requires attention, like following a recipe and measuring out the ingredients into a bowl?


Public Speaking Methods: Pros and Cons
KEY:
Scariness (for speaker)
Boringness (for audience)
Difficulty (amount of prep needed)
Most students
Requires most prep
High
Low
How to improve here?
Best for audience and speaker
METHOD
1 Wing it
2 Talk Through a Set Structure
3A Read Off a Script
3B Just Barely Memorized
3C Happy-Birthday-Level Memorized

Memorization method

	Read out loud from script	Recite without script
Sentence 1	3 times	3 times
Sentence 2	x 3	X 3
Sentences 3,4,5 etc		
Paragraph 2,3,4	Link paras together	Link paras together

- Speed and pausing
- Individual word pronunciation
- Difficult sounds
- Numbers, figures, percentages, acronyms
- Technical terminology
- Intonation of sentences
- Transition from sentence to sentence

- Voice Record and playback
- Video record and playback- feedback from a friend

Eye contact with screen
Notes, script location
Camera background and environment
One take / spliced sections

Good afternoon everyone and thanks for coming to my presentation. Today, I'm going to tell you about a very interesting development in the train industry- hydrogen powered trains. You may not have heard about these before, but within a few years they may become very common and change the face of our industry.

In my talk today, I'm going to outline the problem that hydrogen trains can help to overcome and then go on to look at the technology involved, and the advantages it has over diesel or electric trains. **I'll briefly outline** the business aspects of this development as well.

Ok, to being, here is a photo of a hydrogen-powered train. As you can see, it looks very similar to an ordinary train. The main difference is in the power suppy. This one is called the iLint and is being developed by the French rail manufacturer, Alstrom. Currently Alstrom has orders for 60 hydrogen trains from German rail companies and has contracted to provide maintenance for 30 years. They say that these trains will travel at around d140km/hour and will be able to travel 800km on a single tanks of hydrogen, carrying 300 passengers.

The reason for developing a hydrogen powered train is the need to reduce CO2 emissions from the train sector. In Germany, the target is to reduce CO2 by 40% by 2020 and to use 80% renewable energy by 2050. These are ambitious targets and hydrogen trains are an important part of reaching this aim.

Let me now turn to the technology and how it works. The basic fuel is hydrogen in a fuel cell which is used to generate electricity which powers the main engine. The only emissions are condensed water and steam. Thus, there are zero CO2 emissions in the train operation. Power which is not immediately used is stored in lithium ion batteries, and a converter is used to produced power for the air conditioning, doors, lights and digital displays. The power is managed by a smart power management system, and thus waste energy is minimized.

However, there is one major issue that needs to be considered and that is how hydrogen fuel is produced in the first place. It can be produced by electrolysis, which uses electricity to split water into hydrogen and oxygen, or chemically from natural gas by combining methane and high temperature steam. Both of these methods produce CO2 by the electricity they need. **But,** by using CO2 free wind power to produce electricity, which is then used to produce hydrogen, all CO2 emissions can be avoided.

Right, so we've looked at reasons for using hydrogen to drive trains and the technology behind it. **Now, let's turn our attention to** the business aspects of this development. So, what is the current state of play**? Well,** tests were carried out in 2017 and in early 2018, the first proper network trials will take place in Germany. **Currently** there are 4000 diesel trains running in Germany and Alstrom plans to replace many of these with hydrogen trains.

Another issue is the high cost of electrification of rail lines. In the Uk recently a number of planned electrification schemes were cancelled due to cost issues. **Thus** Alstrom predicts that there will be more demand for non-electric and non-diesel trains in the future. They are hoping that if their hydrogen trains prove to be cost effective and successful that they can supply trains to meet that demand.

OK, so that brings us to the end of my short talk. **I'd now like you to discuss** the following questions which I have prepared about the topic of greener trains in general, and more specifically about hydrogen trains. (612)

Introduce the topic

- *My name is … and I'm going to talk about …*
- *In this presentation, I will talk about/ outline / explain / go through / show*

Outlining

- *I will focus on*
- *Then I'll show*
- *Finally I'll highlight*

Sequencing/Ordering

- *firstly… secondly… thirdly…*
- *then… next… finally/lastly…*
- *let's start with…*
- *let's move/go on to…*
- *now we come to…*
- *that brings us to…*
- *let's leave that…*
- *that covers…*

Contradicting (*to say the opposite*)

- *In fact*
- *actually*

Summarizing

- *to sum up*
- *in brief*
- *in short*

Concluding

- *in conclusion*
- *to conclude*

Highlighting (*to stress*)

- *in particular*
- *especially*

Digressing (*to go off the point*)

- *by the way*
- *let's get back to…*

Giving reasons/causes

- *therefore*
- *so*
- *as a result*
- *that's why*

Contrasting (*difference*)

- *but*
- *however*

Comparing (*similar*)

- *similarly*
- *in the same way*

Giving examples

- *for example*
- *for instance*
- *such as*

Generalizing

- *usually*
- *generally*

Signalling the end

- *That brings me to the end of my presentation.*
- *That completes my presentation.*
- *Before I stop/finish, let me just say…*
- *That covers all I wanted to say today.*

Recommending

- *So, I would suggest that we…*
- *I'd like to propose…(more formal)*
- *In my opinion, the best solution is…*

Summarizing

- *Let me just run over the key points again.*
- *I'll briefly summarize the main issues.*
- *To sum up…*
- *Briefly…*

Closing

- *Thank you for your attention/time.*
- *Thank you for listening.*
- *I hope you will have gained an insight into…*

Concluding

- *As you can see, there are some very good reasons…*
- *In conclusion…*
- *I'd like to leave you with the following thought/idea.*

Inviting questions

- *I'd be glad to try and answer any questions.*
- *Does anyone have any questions?*

Tips for recording yourself reading your presentation.

- Type out parts (or all) of your presentation
- Separate them into lines, like above
- Practice reading at normal speed, focusing on individual sounds, intonation and rhythm.
- Practice difficult sounds or combinations of words again and again.
- Record yourself reading and listen to yourself to identify the problem words
- Come back a day later and record yourself again. By listening to your own voice you will become more aware of it and identify sounds or intonation that you need to work on. This will also help you with speed of delivery and pausing.
- Write down a list of individual words you find a bit tricky to pronounce and practice saying them individually-record yourself doing so.

  e.g. Technical words, words with many syllables, words with sounds which can be tricky for Chinese speakers.

- Write down small phrases or series of words that you find tricky. Record yourself saying them and listen back.

  Eg. Consonant cluster; phrases which are linked together; -ed endings; short words (prepositions)

- Send your sound file to a friend for them to listen and give feedback. And you do the same for them.

problem	cause	effect	solution
Speed of delivery too fast.	Nerves, over reliance on poorly memorized text, reciting not communicating. Conscious mind focusing on recitation, not on content or audience Poor rehearsal and timing planning Trying to cram in too many ideas; failure to identify essential info	Audience lost or lose interest and start thinking- I wonder if I have any Wechat messages Audience have to 'work' too hard to follow the talk and switch off	Either memorize to the 'Happy Birthday' level to free conscious mind to think about ideas, audience, eye contact. Improve selection of content. Plan and rehearse again and again for timing Use pausing to mark changes of points, sections- Count to 3 mentally before moving on. Record yourself

The questioner may not know about your research area, or the relevant literature or important concepts or the procedures you followed.

Anticipate questions you may be asked.
Which aspects of your research might be tricky to follow, confusing? Which are the most relevant and interesting?

A: Reasons for choosing this research topic.

- Where did your research idea come from?
- Is there a 'gap' in research that your study attempts to fill?
- What was your research question?
- Is your research 'original' or 'novel'?

You must 'sell' your research. It must sound interesting, professional, high quality and relevant. You must be enthusiastic about your research and keen to teach others about it.

B: Design / research process

- What was the easiest / most difficult part of the research project?
- What external support did you use during this research process- e.g. tutors, peers, friends, advisors etc.? What support did you get from them?
- Did you use AI to help you with your research? Which AI and how did you use it?
- What were the main study resources you used to learn about your topic for this research- e.g. textbooks, lecture or seminar notes and PPTS, online educational resources (videos, wikis, blogs, discussion forums etc.)? Can you recommend the best educational sources you found.
- Describe a problem you encountered and how you overcame it.
- Did you encounter anything unexpected in your research- i.e. quantitative or qualitative results or outcomes that were different from what you expected?
- If you were starting your research process again what would you do differently the 2nd time round and why?

C: Key terminology, concepts, ideas needed to understand your research.

- Explain 3 key terms / concepts that are important to your research
- Why are they significant / relevant to your research?

D: Detailed aspects of your research

1. Introduction- Motivation + Aims and Objectives
2. Literature Review and related work
3. Methodology and Study Design
4. Experimental Results, Implementation and Evaluation.
5. Discussion and Conclusion

- Summarize your research and findings in 1 minute / 3 minutes.
- Which papers from the literature were most useful or influential for your research and why?
- Did you come across any bad or dubious research in your literature review? What criteria did you use to include (or exclude) research papers in your literature review?
- What methodology did you use for this research and why?
- Was your research mostly quantitative or qualitative?
- Which field of study is your project related to- e.g. psychology, biology, engineering, medicine, technology etc.? You are not an expert in this area. How did this impact your research?
- Briefly explain an interesting technical aspect of the experimental or implementation part of your research.
- If you designed a product or your research is designed to improve a product- describe the product and how it can be improved.
- Do you think your research findings are replicable? If someone else repeated your experimental research would you get the same findings?
- Summarize your final discussion and relate it to your original research question.

D: Linking this research / product to further work / external markets

- Who would be interested in using or citing your research? Other researchers, businesses, industry, universities, individuals?
- What other fields of study does it relate to?
- What other researchers or designers might learn something useful from your research?
- Does your research have potential commercial applications? Have you researched the market in this area?

D: Problems in your work- A reviewer notices a problem in your work or asks you about choices you made. Can you think of examples of these choices from your work?

- You didn’t do X or Y- why not?
- You didn’t use X or Y- why not?
- Why didn’t you use X instead of Y?
- Do you think X would have given you’re a better results?

COMPUTER SCIENCE DISSERTATION QUESTION AND ANSWER SESSION

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- You didn't use X or Y- why not?
- Why didn't you use X instead of Y?
- Do you think X would have given you're a better results?
- Are you aware of the recent research by….?
- Could you say how you used the research by X in your project?